

Image Manipulation

Image Filtering: Change range (brightness)

$$g(x, y) = T_r(f(x, y))$$



Image Manipulation

Image Filtering: Change range (brightness)

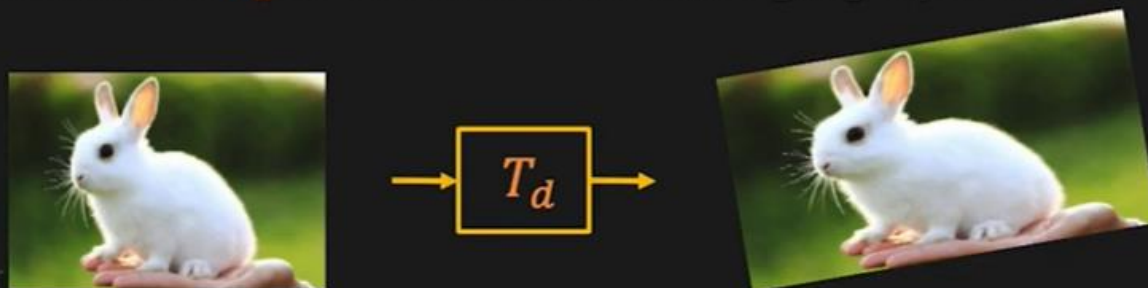
$$g(x, y) = T_r(f(x, y))$$



Image Warping: Change domain (location)

$$g(x, y) = f(T_d(x, y))$$

Transformation T_d is a coordinate changing operator



Global Warping/Transformation



Translation



Rotation



Scaling and Aspect

$$g(x, y) = f(T(x, y))$$



Affine



Projective



Barrel

Transformation T is the same over entire domain

Often can be described by just a few parameters

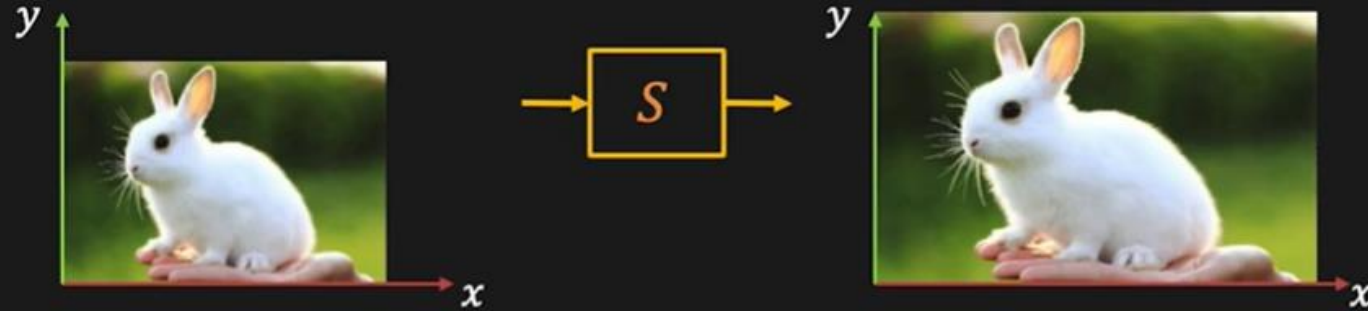
2x2 Linear Transformations



T can be represented by a matrix.

$$\mathbf{p}_2 = T\mathbf{p}_1 \quad \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = T \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \quad \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

Scaling (Stretching or Squishing)

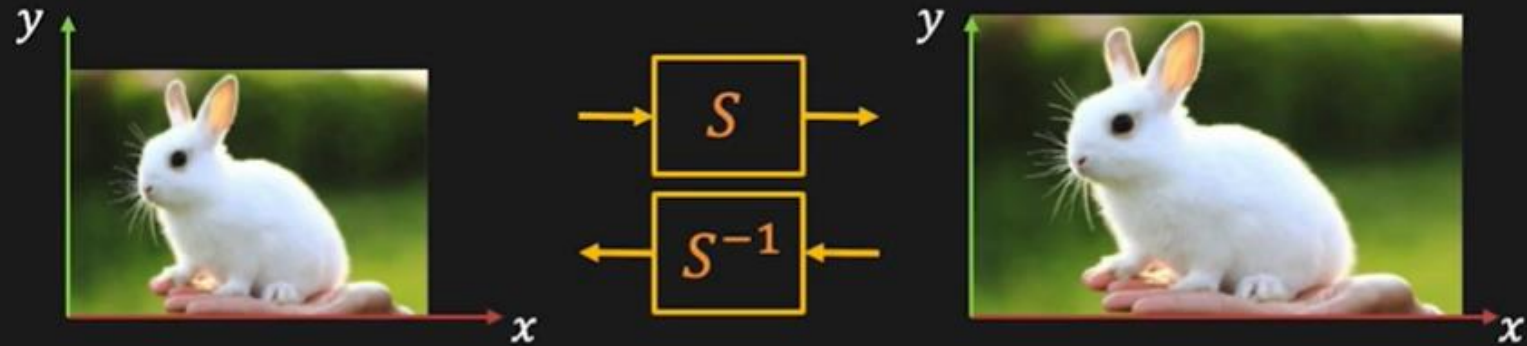


Forward:

$$x_2 = ax_1 \quad y_2 = by_1$$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = S \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

Scaling (Stretching or Squishing)



Forward:

$$x_2 = ax_1 \quad y_2 = by_1$$

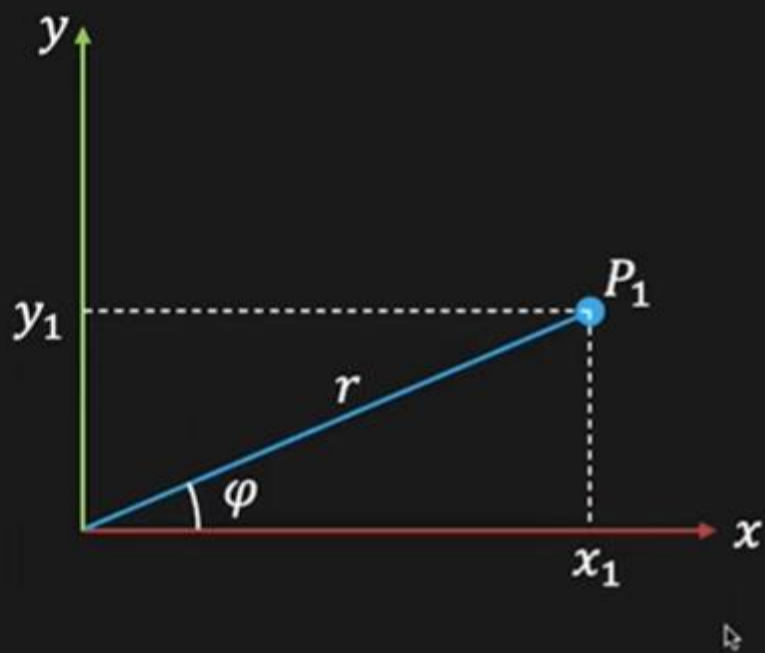
$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = S \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

Inverse:

$$x_1 = \frac{1}{a}x_2 \quad y_1 = \frac{1}{b}y_2$$

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = S^{-1} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1/a & 0 \\ 0 & 1/b \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

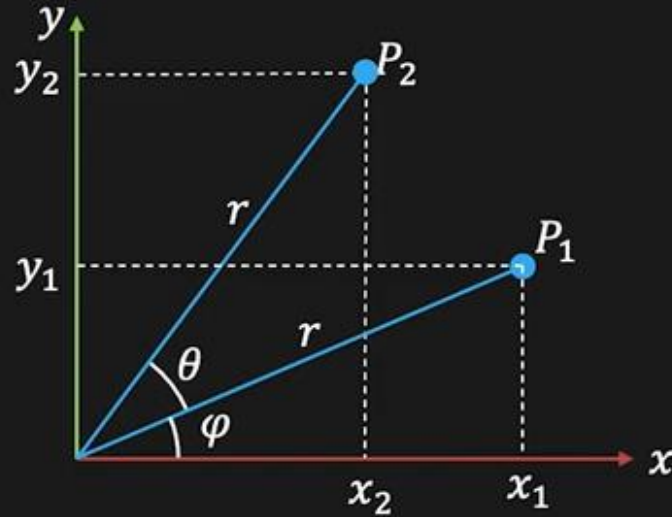
2D Rotation



$$x_1 = r \cos(\varphi)$$

$$y_1 = r \sin(\varphi)$$

2D Rotation



$$x_1 = r \cos(\varphi)$$

$$y_1 = r \sin(\varphi)$$

$$x_2 = r \cos(\varphi + \theta)$$

$$x_2 = r \cos \varphi \cos \theta - r \sin \varphi \sin \theta$$

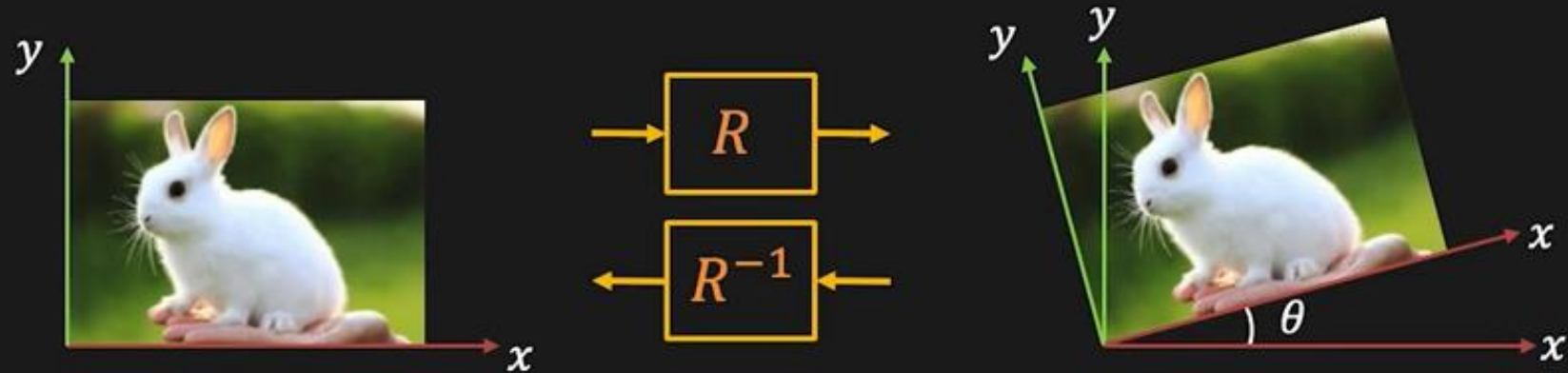
$$x_2 = x_1 \cos \theta - y_1 \sin \theta$$

$$y_2 = r \sin(\varphi + \theta)$$

$$y_2 = r \cos \varphi \sin \theta + r \sin \varphi \cos \theta$$

$$y_2 = x_1 \sin \theta + y_1 \cos \theta$$

Rotation



Forward:

$$x_2 = x_1 \cos \theta - y_1 \sin \theta$$

$$y_2 = x_1 \sin \theta + y_1 \cos \theta$$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = R \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

Inverse:

$$x_1 = x_2 \cos \theta + y_2 \sin \theta$$

$$y_1 = -x_2 \sin \theta + y_2 \cos \theta$$

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = R^{-1} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

Skew



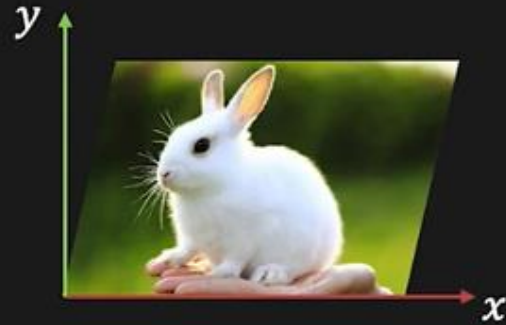
Horizontal Skew:

$$x_2 = x_1 + m_x y_1$$

$$y_2 = y_1$$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = S_x \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 1 & m_x \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

Skew



Horizontal Skew:

$$x_2 = x_1 + m_x y_1$$

$$y_2 = y_1$$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = S_x \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 1 & m_x \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$



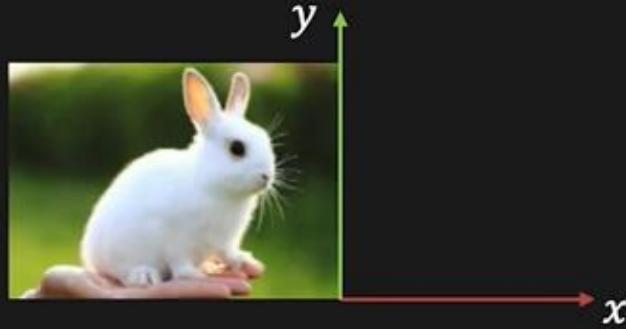
Vertical Skew:

$$x_2 = x_1$$

$$y_2 = m_y x_1 + y_1$$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = S_y \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ m_y & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

Mirror



Mirror about Y-axis:

$$x_2 = -x_1$$

$$y_2 = y_1$$

$$M_y = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Mirror about line $y = x$:

$$x_2 = y_1$$

$$y_2 = x_1$$

$$M_{xy} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

2x2 Matrix Transformations

Any transformation of the form:

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

- Origin maps to the origin
- Lines map to lines
- Parallel lines remain parallel
- Closed under composition

$$\left. \begin{array}{l} \mathbf{p}_2 = T_{21}\mathbf{p}_1 \\ \mathbf{p}_3 = T_{32}\mathbf{p}_2 \\ \mathbf{p}_3 = T_{31}\mathbf{p}_1 \end{array} \right\} \mathbf{p}_3 = T_{32}\mathbf{p}_2 = T_{32}T_{21}\mathbf{p}_1 \Rightarrow T_{31} = T_{32}T_{21}$$